Quiz 4

Variance and discrete joint/conditional laws

Try each question before reading the worked solutions on page 2.

1. $\mathbb{E}[X] = 2$ and $\mathbb{E}[X^2] = 7$. What is $\mathrm{Var}(X)$?

A. 5

B. 3

C. 7

D. -3

2. $\mathrm{Var}(X) = 3$. What is $\mathrm{Var}(5 - 2X)$?

A. -6

B. -1

C. 6

D. 12

3. For a joint PMF $p_{X,Y}(x, y)$, how is $p_X(x)$ obtained?

A. Sum over y, holding x fixed.

B. Sum over x, holding y fixed.

C. Divide every entry by its row total.

D. Use the largest entry in each row.

4. $P(X = 1, Y = 2) = 0.12$ and $P(X = 1) = 0.30$. What is $P(Y = 2 \mid X = 1)$?

A. 0.036

B. 0.12

C. 0.40

D. 2.5

5. To establish independence for a finite joint table, which test is sufficient?

A. One cell equals its marginal product.

B. The two marginals are identical.

C. The support is rectangular.

D. Every cell equals its marginal product.

Worked solutions

1. Variance from moments

Answer B

Variance measures squared deviations from the mean. Expanding $\mathbb{E}[(X - \mathbb{E}[X])^2]$ gives the useful identity

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 7 - 2^2 = 3.$$

The second moment $\mathbb{E}[X^2]$ and the squared mean $(\mathbb{E}[X])^2$ are different quantities.

2. Shifts and scaling

Answer D

Adding a constant shifts all values equally, so it does not change their spread. Multiplying by b multiplies squared deviations by b^2:

$$\mathrm{Var}(a + bX) = b^2 \,\mathrm{Var}(X), \qquad \mathrm{Var}(5 - 2X) = (-2)^2(3) = 12.$$

The negative coefficient is squared; a variance cannot be negative.

3. Marginalizing a joint PMF

Answer A

For a fixed x, the events $\{X = x, Y = y\}$ over all possible y are disjoint and together form $\{X = x\}$. Therefore

$$p_X(x) = P(X = x) = \sum_{y \in S_Y} P(X = x, Y = y) = \sum_{y \in S_Y} p_{X,Y}(x, y).$$

Joint probabilities outside the joint support contribute zero.

4. Conditioning on a discrete value

Answer C

The condition is $X = 1$, so its probability is the denominator. Since $P(X = 1) = 0.30 > 0$, the conditional-probability formula gives

$$P(Y = 2 \mid X = 1) = \frac{P(X = 1, Y = 2)}{P(X = 1)} = \frac{0.12}{0.30} = 0.40.$$

This rescales the probabilities within the event $\{X = 1\}$.

5. Checking independence

Answer D

Independence requires the joint PMF to factor into its marginals at every pair of possible values:

$$p_{X,Y}(x, y) = p_X(x) p_Y(y) \quad \text{for all } x \in S_X,\ y \in S_Y.$$

Check every cell, including zero entries. One matching cell, equal marginals, or a rectangular support alone does not establish independence.